

PROJECT DOCUMENTATION

HeritageLens

AI-Powered Indian Art and Cultural Heritage Recognition System

Computer Vision | Convolutional Neural Network | Cultural Heritage

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Abstract

HeritageLens is an AI-powered web application designed to recognize traditional Indian art forms from uploaded images and to present structured cultural information about the recognized art. The system applies Artificial Intelligence, Machine Learning, and Computer Vision, using a Convolutional Neural Network (CNN) built through transfer learning on the MobileNetV2 architecture, to classify an artwork into one of ten supported categories: Madhubani, Warli, Kalamkari, Pattachitra, Gond Art, Tanjore Painting, Phad Painting, Cheriya Painting, Kerala Mural, and Mandana Art.

For every recognized artwork, the application reports the detected art form, a confidence percentage, and the next most likely candidates, together with its origin, history, cultural significance, characteristics, materials, traditional techniques, and interesting facts. Beyond recognition, the platform offers an Explore Heritage gallery with search and filters, a Dynasties and Heritage section covering the Maurya, Gupta, Pallava, Chalukya, Rashtrakuta, and Chola dynasties, a Learn and Discover section, and a locally stored recognition history.

The solution follows a modular architecture. The frontend is built with HTML, CSS, and vanilla JavaScript; the backend is a Python Flask REST API that uses Pillow and NumPy for image preprocessing; the AI component uses TensorFlow and Keras; and a JSON-based knowledge base stores all cultural information. This report documents the problem, objectives, cultural background, system analysis and design, algorithms, implementation, output, testing, limitations, and future scope of the project.

Keywords: Indian Art Recognition, Cultural Heritage, Convolutional Neural Network, MobileNetV2, Transfer Learning, Flask REST API, Computer Vision, Dynasties of India.

List of Abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
API	Application Programming Interface
REST	Representational State Transfer
JSON	JavaScript Object Notation
HTTP	Hypertext Transfer Protocol
HTML / CSS	Hypertext Markup Language / Cascading Style Sheets
UI / UX	User Interface / User Experience
ReLU	Rectified Linear Unit
PIL	Python Imaging Library (Pillow)
UNESCO	United Nations Educational, Scientific and Cultural Organization

1. Introduction

1.1 Background

India possesses one of the world's oldest and most diverse artistic traditions. Across its states and regions, communities have preserved paintings, scroll narratives, wall and floor decorations, textile arts, and temple murals that carry religious belief, folklore, seasonal ritual, and community identity. Many of these traditions were passed from one generation to the next through practice rather than through formal documentation, which makes them both valuable and vulnerable.

In the present day, younger generations and visitors often encounter these art forms without the context needed to understand them. A viewer may admire a Warli painting or a Tanjore panel without knowing where it originated, what materials were used, or what it represents. Digital technology offers an opportunity to close this gap by making cultural knowledge instantly accessible.

1.2 About HeritageLens

HeritageLens is an AI-powered cultural recognition and learning platform. A user uploads an image of an Indian traditional artwork, the system analyses it with a trained image classification model, and the result is presented with a confidence score and a detailed cultural profile. The platform is organized into six navigation sections: Home, Recognize Art, Explore Heritage, Dynasties and Heritage, Learn and Discover, and About.

1.3 Motivation

The project is motivated by the need to make cultural learning engaging and reachable. Recognition-based interaction converts a passive act, looking at an artwork, into an active learning experience: the user asks "what is this?" and receives a meaningful, structured answer. The addition of the Dynasties and Heritage section extends the same idea from art to the historical rulers whose patronage shaped India's architecture and artistic life.

1.4 Importance of Preserving Cultural Heritage

Cultural heritage is a record of a society's identity, values, and creativity. Traditional art forms are living heritage: they depend on artisans, materials, and techniques that can fade if they are not documented, taught, and appreciated. Digital documentation does not replace the practice of these arts, but it broadens awareness, supports education, and encourages appreciation among people who might otherwise never encounter them.

1.5 Role of Artificial Intelligence

Computer Vision allows software to interpret visual patterns. Traditional Indian art forms are strongly recognizable by their visual signatures, such as the white geometric figures of Warli, the densely filled line work of Madhubani, or the gold foil of Tanjore paintings. A CNN can learn these signatures from labelled examples and apply them to new images, which makes image classification a natural fit for cultural recognition.

2. Problem Statement, Objectives and Scope

2.1 Problem Statement

India has numerous traditional paintings, folk art forms, and regional traditions, but awareness of their origins, significance, and techniques is limited, especially among younger people. No simple digital tool lets an ordinary user identify a traditional Indian art form from an image and immediately learn reliable cultural information about it. An interactive platform that uses modern AI to recognize traditional art and support cultural learning is therefore needed.

2.2 Objectives

1. To build an image classification system that recognizes traditional Indian art forms using deep learning.
2. To display the predicted art form with a confidence percentage and the other most likely candidates.
3. To provide structured cultural information for each art form: origin, history, significance, characteristics, materials, techniques, and facts.
4. To offer an Explore Heritage gallery with search and filtering by region and technique.
5. To provide a Dynasties and Heritage section, a local recognition history, and an educational Learn and Discover section.
6. To design a modular, scalable system to which new art forms and improved models can be added.

2.3 Scope of the Project

- **Recognition:** ten traditional art forms, using JPG, JPEG, or PNG images of up to 5 MB.
- **Knowledge base:** structured cultural information for each supported art form, stored in JSON.
- **Exploration:** searchable galleries for art forms and for dynasties, with region, technique, and era filters.
- **Education:** cultural facts, technique explanations, and short articles in the Learn and Discover section.
- **Out of scope:** art forms outside the trained categories, identification of artists, and valuation of artworks.

2.4 Target Users

Table 1: Target users and benefits

User group	How HeritageLens helps
Students and teachers	A visual, interactive resource for lessons on art, history, and culture

User group	How HeritageLens helps
Tourists	Quick identification of art seen at museums, markets, and heritage sites
Art enthusiasts and researchers	Deeper understanding of materials and techniques; a structured reference
Cultural organizations	A digital platform that promotes traditional art forms

3. Cultural Background: Indian Traditional Art Forms

The recognition model supports ten traditional art forms. Each has a distinct region of origin, medium, and visual style. The summaries below describe the general characteristics that make these art forms visually distinguishable, which is the basis for automated recognition.

Table 2: Supported art forms

Art form	Region	Distinguishing traits
Madhubani	Bihar (Mithila)	Dense, intricate line work; bold borders; deities, nature, ritual scenes
Warli	Maharashtra	White geometric figures (circles, triangles, squares) on an earthen background
Kalamkari	Andhra Pradesh	Hand-painted or block-printed cotton textiles; epics and floral vines
Pattachitra	Odisha, West Bengal	Cloth-based scroll paintings; bold outlines; Puranic and Jagannath themes
Gond Art	Madhya Pradesh	Animals and trees filled with dots, dashes, and flowing lines
Tanjore Painting	Tamil Nadu	Devotional panels with gold foil, gems, and rounded divine figures
Phad Painting	Rajasthan	Long cloth scrolls narrating folk epics and local heroes
Cheriyal Painting	Telangana	Narrative scroll paintings with vivid colours, often on a red background
Kerala Mural	Kerala	Temple wall paintings using a five-colour palette and expressive figures
Mandana Art	Rajasthan, Madhya Pradesh	Ritual wall and floor designs in white on a red or earthen base

3.1 Madhubani

Madhubani, also known as Mithila painting, originates in the Mithila region of Bihar. It is recognized by its densely filled compositions, decorative borders, and depictions of deities, nature, and ritual scenes. Traditional practice uses natural pigments and simple tools such as twigs, brushes, and fingers.

3.2 Warli

Warli art is practised by the Warli tribe of the Sahyadri range in Maharashtra. It was originally drawn on the mud walls of huts during harvest and weddings. Human figures are built from triangles, and circular dance scenes (tarpa) express the community's bond with nature and the cycle of farming and celebration. Traditional materials include rice paste, mud walls, and a bamboo stick.

3.3 Kalamkari

Kalamkari, from Andhra Pradesh, is a textile art in which cotton cloth is hand-painted or block-printed. The name refers to work done with a pen (kalam). Themes include episodes from the epics and floral vines, and natural dyes are traditionally used.

3.4 Pattachitra

Pattachitra is a cloth-based scroll painting tradition of Odisha and West Bengal. It is known for bold outlines, rich natural colours, and narratives drawn from Puranic stories and the worship of Lord Jagannath.

3.5 Gond Art

Gond art belongs to the Gond community of Madhya Pradesh. Its animals and trees are filled with patterns of dots, dashes, and flowing lines, giving the paintings a lively sense of movement.

3.6 Tanjore Painting

Tanjore, or Thanjavur, painting of Tamil Nadu produces rich devotional panels. Its recognizable features are gold foil, inlaid gems, and rounded, serene figures of deities set within ornate frames.

3.7 Phad Painting

Phad painting of Rajasthan is executed on long cloth scrolls that narrate the stories of local folk heroes and deities. Historically, the scrolls were used in performances in which a narrator explained the story to an audience.

3.8 Cheriyal Painting

Cheriyal paintings of Telangana are narrative scrolls that portray stories in a sequence of panels. They are noted for vivid colours and a characteristic red background.

3.9 Kerala Mural

Kerala murals are wall paintings found in temples and palaces of Kerala. They use a restricted, symbolic five-colour palette and depict mythological figures with expressive features and elaborate ornamentation.

3.10 Mandana Art

Mandana is a folk art of Rajasthan and Madhya Pradesh in which geometric and symbolic designs are drawn on walls and floors, typically in white on a red or earthen surface, for festivals and household rituals.

4. Dynasties and Heritage

4.1 Purpose of the Module

India's artistic traditions developed alongside a long history of dynasties whose rulers commissioned temples, monuments, sculptures, and inscriptions. To help users connect art with history, HeritageLens includes a dedicated Dynasties and Heritage section. It presents major dynasties as illustrated cards showing the dynasty name, the period, and the region it governed, with a search box and an era filter to help users find what they are looking for.

4.2 Dynasties Covered

Table 3: Dynasties in the Dynasties and Heritage section

Dynasty	Period	Region	Heritage highlights
Maurya	c. 322 BCE – 185 BCE	Most of the Indian subcontinent	Ashokan edicts and pillars; the Sanchi Stupa
Gupta	c. 320 CE – 550 CE	Northern and central India	Early structural temples; advances in learning and literature
Pallava	c. 275 CE – 897 CE	Northern Tamil Nadu and Andhra coast	Mahabalipuram rock-cut and structural temples
Chalukya	c. 543 CE – 1200 CE (two main phases)	Deccan region (modern Karnataka and parts of Maharashtra, Andhra Pradesh)	Temples of Badami, Aihole, and Pattadakal
Rashtrakuta	c. 753 CE – 982 CE	Deccan plateau (modern Maharashtra, Karnataka)	Kailasa Temple at Ellora
Chola	c. 850 CE – 1279 CE (imperial period)	Tamil Nadu, with influence across southern India, Sri Lanka, and maritime Southeast Asia	Brihadisvara Temple; Chola bronzes

4.3 Maurya Dynasty

The Maurya Empire, founded by Chandragupta Maurya, unified much of the Indian subcontinent under a single administration with its capital at Pataliputra. Emperor Ashoka is remembered for inscribing edicts on rocks and polished stone pillars and for his patronage of Buddhism. Monuments associated with this period, such as the Great Stupa at Sanchi, and the sculpted pillar capitals, remain important symbols of early Indian art and architecture.

4.4 Gupta Dynasty

The Gupta period is widely regarded as a classical age of Indian culture. It saw notable achievements in mathematics, astronomy, literature, and sculpture. Early structural temples, including the brick and stone temples of northern and central India, were built during this era and influenced later temple architecture.

4.5 Pallava Dynasty

The Pallavas ruled from Kanchipuram and are known for a major transition in South Indian architecture from rock-cut caves to structural stone temples. The monuments at Mahabalipuram, including the Shore Temple and the monolithic rathas, are among the finest surviving examples and form a UNESCO World Heritage Site. Pallava art and script also influenced cultures in Southeast Asia.

4.6 Chalukya Dynasty

The Chalukyas ruled the Deccan in two main phases, first from Badami and later from Kalyani. Their architectural experiments at Badami, Aihole, and Pattadakal combined northern and southern temple styles, and Pattadakal is a UNESCO World Heritage Site. These sites are often studied as a laboratory of early temple design.

4.7 Rashtrakuta Dynasty

The Rashtrakutas governed the Deccan plateau from Manyakheta. Their most celebrated monument is the Kailasa Temple at Ellora, a temple carved from a single rock, which demonstrates extraordinary engineering and sculptural skill. The dynasty also patronized literature, including works in Kannada.

4.8 Chola Dynasty

The Cholas rose to imperial power in the ninth century and, under rulers such as Rajaraja I and Rajendra I, extended their influence across southern India, Sri Lanka, and maritime Southeast Asia. The Brihadisvara Temple at Thanjavur, completed around 1010 CE, is a masterpiece of Dravidian architecture, and the Great Living Chola Temples are recognized by UNESCO. The Cholas are also famous for bronze sculpture, particularly images of Nataraja, made by the lost-wax method. The Tanjore art tradition included in the recognition model shares its home region with this heritage.

4.9 Linking Dynasties, Architecture, and Art

Dynastic patronage created the conditions in which art and architecture flourished. Temples provided walls for murals and sculpture, courts supported craftspeople and scholars, and trade routes carried styles across regions. Presenting dynasties next to art forms helps users see that India's heritage is a connected story of rulers, regions, materials, and skills rather than a set of isolated objects.

4.10 Implementation of the Module

- Dynasty information is maintained in the JSON-based knowledge base, in the same way as art form information, so new dynasties can be added without changing the code.
- The frontend renders each dynasty as a card with an image, name, period, and region.

- A search box (dynasty, temple, region) and an era drop-down filter narrow the displayed cards.

5. System Analysis and Requirements

5.1 Existing Situation

At present, users who wish to identify an Indian art form must rely on manual web searches, books, or expert knowledge. Text-based search requires the user to already know the name of the art form, while image-based search engines return general visual matches rather than structured cultural explanations. As a result, there is no dedicated and beginner-friendly tool for cultural recognition.

5.2 Proposed System

The proposed system accepts an image as input and returns the recognized art form along with a structured cultural profile. It combines a trained CNN classifier, a REST API, a knowledge base, and an interactive interface, so that recognition and learning occur in one place.

5.3 Functional Requirements

Table 4: Functional requirements

ID	Requirement
FR1	The system shall allow the user to upload a JPG, JPEG, or PNG image by drag-and-drop or file selection.
FR2	The system shall reject unsupported file types and images larger than 5 MB with a clear error message.
FR3	The system shall classify the image and display the detected art form and confidence percentage.
FR4	The system shall display the other most likely candidates with their percentages.
FR5	The system shall display origin, history, significance, characteristics, materials, techniques, and facts.
FR6	The system shall provide a searchable Explore Heritage gallery with region and technique filters.
FR7	The system shall provide a Dynasties and Heritage section with search and era filters.
FR8	The system shall store and display recent recognitions locally in the browser.
FR9	The system shall indicate whether it is running in demo or real model mode.

5.4 Non-Functional Requirements

- **Usability:** a simple, clear interface with a visible progress indicator during analysis.
- **Performance:** a lightweight model (MobileNetV2) for fast inference on ordinary hardware.
- **Responsiveness:** a layout that adapts to desktop, tablet, and mobile screens.
- **Maintainability:** separate frontend, backend, model, and data components.
- **Scalability:** new art forms and dynasties are added through data and retraining, not code changes.
- **Reliability and honesty:** demo predictions are always labelled as demo output.

5.5 Hardware and Software Requirements

Table 5: Hardware and software requirements

Item	Requirement
Processor and memory	A standard modern computer; a GPU is optional and useful only for training the model
Operating system	Windows, macOS, or Linux
Programming language	Python 3
Libraries	Flask, TensorFlow/Keras, Pillow, NumPy
Client	Any modern web browser (Chrome, Edge, Firefox, Safari)

5.6 Feasibility Study

Technical feasibility: all components use mature, freely available technologies, and transfer learning reduces the need for large datasets and computing power. **Economic feasibility:** the project uses open-source software and can run on modest hardware, so its cost is low. **Operational feasibility:** the interface requires no technical knowledge, and the workflow of uploading an image and reading a result is intuitive.

6. System Architecture and Design

6.1 Layered Architecture

HeritageLens is organized into loosely coupled layers. The frontend communicates with the backend only through the REST API, and the backend communicates with the model and the knowledge base through separate modules. This separation allows any layer to be modified or replaced independently.

```
Frontend (HTML / CSS / JavaScript)
|   HTTP requests (JSON, multipart/form-data)
v
Backend API (Flask)
|-- Image preprocessing (Pillow, NumPy)
|-- AI/ML model (TensorFlow/Keras, MobileNetV2)
`-- Cultural knowledge base (JSON)
|
v
Prediction + confidence + cultural information -> Result display
```

6.2 Recognition Data Flow

1. The user selects or drops an image on the Recognize Art page.
2. The frontend checks the file type and size and sends the image to `/api/predict` as `multipart/form-data`.
3. The backend opens the image with Pillow, converts it to RGB, resizes it to the model input size, and converts it to a NumPy array.
4. The CNN produces a probability for each art class through a softmax layer.
5. The class with the highest probability becomes the detected art form; its probability is the confidence, and the next highest classes are listed as other candidates.
6. The backend attaches the matching entry from the knowledge base and returns everything as JSON.
7. The frontend displays the result and stores it in the browser's local history.

6.3 Module Design

Table 6: System modules

Module	Location	Responsibility
User interface	frontend/index.html, style.css	Page structure, styling, responsive layout, animations
Client logic	frontend/app.js	Section navigation, cards, search and filters, upload, API calls, history
Web server and API	backend/app.py	Serves the frontend; exposes the REST endpoints; validates uploads
Predictor	model/predictor.py	Image preprocessing, model loading, inference, demo mode

Module	Location	Responsibility
Knowledge base	data/art_forms.json	Cultural information for art forms (and dynasty data)

6.4 Knowledge Base Design

The knowledge base is a JSON file. Each art form is an object whose fields correspond to the information shown to the user. The structure below is illustrative; field names in the implementation may differ slightly.

```
{
  "name": "Warli",
  "origin": "Maharashtra",
  "history": "Practised by the Warli tribe of the Sahyadri range ...",
  "significance": "Expresses a community's bond with nature ...",
  "characteristics": ["Stick-figure humans from triangles", "..."],
  "materials": ["Rice paste", "Mud walls", "Bamboo stick"],
  "techniques": ["Geometric shapes only", "Ochre and red mud base"],
  "facts": ["The sun and moon are drawn as circles."]
}
```

JSON was chosen over a SQL database because the data is small, mostly read-only, and easy to edit. If the content or number of users grows, the same structure can be moved to SQLite or another database with little change to the rest of the system.

6.5 Project Structure

```
HeritageLens/
|-- frontend/      index.html, style.css, app.js
|-- backend/       app.py
|-- model/          predictor.py (and the trained model file)
|-- data/           art_forms.json (knowledge base)
|-- requirements.txt
`-- README.md
```

7. Technology Stack

Table 7: Technology stack

Layer	Technology	Purpose
Frontend	HTML	Structure of pages and the five main sections
Frontend	CSS	Layout, colours, responsive design, animations
Frontend	Vanilla JavaScript	Interaction, API calls, search and filters, drag-and-drop
Frontend	localStorage	Browser-based storage of recognition history
Backend	Python and Flask	Web server and REST API framework
Backend	Pillow (PIL)	Opening, resizing, and converting uploaded images
Backend	NumPy	Numerical operations on image arrays and probabilities
AI / ML	TensorFlow / Keras	Deep learning framework for building and running the model
AI / ML	MobileNetV2	Pretrained CNN used as the base model (transfer learning)
Data	JSON file	Cultural knowledge base; no SQL database required

7.1 Frontend

The frontend uses plain HTML, CSS, and JavaScript without a framework. This keeps the project simple and easy to understand. JavaScript handles navigation between sections without page reloads, renders art and dynasty cards from JSON data, and sends the image to the backend using the Fetch API and FormData.

7.2 Backend

Flask is a lightweight Python web framework. It serves the static frontend files and provides the API endpoints. Because the AI model is also written in Python, the backend can call it directly without an intermediate service.

7.3 AI and Machine Learning

TensorFlow with the Keras interface is used to define, train, and run the CNN. MobileNetV2 is used as the base network because it is compact and fast while providing strong general image features.

7.4 Data Storage

All cultural content is stored in a JSON knowledge base. The recognition history is stored separately in the browser's localStorage, which keeps personal history on the user's own device.

8. Algorithms Used

8.1 Convolutional Neural Network (CNN)

A Convolutional Neural Network is a deep learning architecture designed for grid-like data such as images. Instead of treating each pixel independently, a CNN applies small learnable filters that slide across the image and respond to local patterns. Early layers detect simple features such as edges and colour transitions; deeper layers combine them into textures, shapes, and motifs; and the final layers use these features to decide the class of the image.

Table 8: Main components of a CNN

CNN component	Function
Convolutional layer	Applies learnable filters to detect local features such as edges, lines, and textures
Activation (ReLU)	Introduces non-linearity so that complex patterns can be learned
Pooling	Reduces the spatial size of feature maps while keeping the strongest responses
Fully connected (dense) layer	Combines extracted features to produce class scores
Softmax	Converts class scores into probabilities that sum to one

This approach suits Indian art recognition because each art form has distinctive visual signatures: Warli uses white geometric figures on an earthen background, Madhubani uses intricate line work and fills, Gond art uses dots and dashes, and Tanjore paintings use gold foil and rich colour.

8.2 Transfer Learning

Training a deep CNN from scratch requires a very large labelled dataset and considerable computing power. Transfer learning avoids this by starting from a network that has already learned general visual features from a large image collection. The pretrained layers are reused as a feature extractor, and only a new classification head is trained for the Indian art categories. This gives good results with a much smaller dataset and shorter training time.

8.3 MobileNetV2

MobileNetV2 is an efficient CNN architecture designed for devices with limited resources. It relies on depthwise separable convolutions, which split a standard convolution into a per-channel filtering step and a channel-combining step, greatly reducing computation. It also uses inverted residual blocks with linear bottlenecks. Its small size and speed make it suitable for a web application that must respond quickly on ordinary hardware.

8.4 Model Structure

Table 9: Layers of the classification model

Layer / component	Role in HeritageLens
RandomFlip, RandomRotation	Data augmentation that increases variety in the training images
Rescaling	Normalizes pixel values to the range expected by the base model
MobileNetV2 (pretrained base)	Extracts general visual features from the image
GlobalAveragePooling2D	Condenses feature maps into a compact feature vector
Dropout	Randomly disables units during training to reduce overfitting
Dense (softmax)	Outputs one probability for each supported art form

8.5 Training Configuration

- **Optimizer:** Adam, an adaptive method that adjusts the learning rate for each parameter and generally converges quickly and stably.
- **Loss function:** sparse categorical cross-entropy, appropriate for multi-class classification with integer class labels.
- **Dataset:** labelled images of Indian traditional artwork, organized into one folder per art class.
- **Regularization:** data augmentation, dropout, and the use of pretrained features.

8.6 Prediction, Confidence, and Candidates

For each uploaded image, the model outputs a probability for every class. The class with the highest probability is reported as the detected art form, and that probability is displayed as the confidence percentage. The next highest classes are listed as other candidates, which helps users judge how certain the prediction is. For example, an image may be classified as Warli with 61.1% confidence, with Mandana Art and Kalamkari shown as alternative candidates.

8.7 Evaluation Measures

The performance of a classifier should be measured on images that were not used in training. Standard measures include accuracy (the proportion of correct predictions), precision and recall for each class, the F1-score, and a confusion matrix, which reveals pairs of art forms that the model tends to confuse. Reporting these measures for the trained model is recommended before the system is presented as a reliable recognizer.

9. Implementation

9.1 Frontend Implementation

The interface uses a deep navy background with golden accents and subtle dotted patterns, inspired by traditional Indian visual styles. The navigation bar provides access to Home, Recognize Art, Explore Heritage, Dynasties and Heritage, Learn and Discover, and About. Cards, buttons, and progress indicators use hover effects and transitions, and the layout adapts to different screen sizes.

9.2 Recognition Workflow in the Interface

On the Recognize Art page, the user drops an image into the upload area or selects a file. After pressing Analyze image, four progress stages are highlighted in turn: Uploading, Preprocessing, Running model, and Fetching cultural notes. When the response arrives, the result panel shows the detected art form, the confidence bar, the other candidates, and the cultural sections.

9.3 REST API

Table 10: API endpoints

Endpoint	Method	Description
/	GET	Serves the frontend (index.html)
/api/arts	GET	Returns the cultural knowledge base as JSON
/api/status	GET	Reports whether the model runs in demo or real mode
/api/predict	POST	Accepts an image and returns the prediction, confidence, and cultural information

The upload is sent as multipart/form-data. The following simplified examples show the client and server sides of the prediction request.

```
// Client (JavaScript)
const formData = new FormData();
formData.append("image", file);
const res = await fetch("/api/predict", { method: "POST", body: formData });
const result = await res.json();

# Server (Flask)
@app.route("/api/predict", methods=["POST"])
def predict():
    file = request.files["image"]
    # validate type and size, preprocess, run model, attach cultural info
    return jsonify(result)
```

9.4 Image Preprocessing

The predictor module uses Pillow to open the uploaded image, convert it to RGB, and resize it to the input size of the model. NumPy then converts the image into an array and adds the batch

dimension expected by the network. Consistent preprocessing during training and prediction is essential for accurate results.

9.5 Recognition History

Each completed recognition is stored in the browser's localStorage, including the art form name, confidence, and a thumbnail. The Recent recognitions area of the Recognize Art page displays these entries as cards. Because the data is stored locally, no user account or database is required, and the history remains private to the user's device.

9.6 Demo Mode and Model Integration

The system never presents fabricated predictions as real machine learning output. When no trained model file is present, the application runs in demo mode and clearly labels its results as demo predictions. The `/api/status` endpoint tells the frontend which mode is active. To activate real recognition, a trained model is saved in Keras format, placed in the location expected by the predictor module with the class order matching the training labels, and the server is restarted.

9.7 Search and Filtering

The Explore Heritage page provides a search box together with region and technique dropdowns. The Dynasties and Heritage page provides a search box and an era filter. Filtering is performed in the browser on the JSON data that has already been loaded, so results update instantly.

10. Application Output

This chapter presents screenshots of the working application and explains each screen.

10.1 Home Page

The home page presents the headline "Discover India's Heritage Through AI", a short description of the platform, and two call-to-action buttons, Recognize Artwork and Explore Heritage. A heritage photograph forms the backdrop of the hero section.

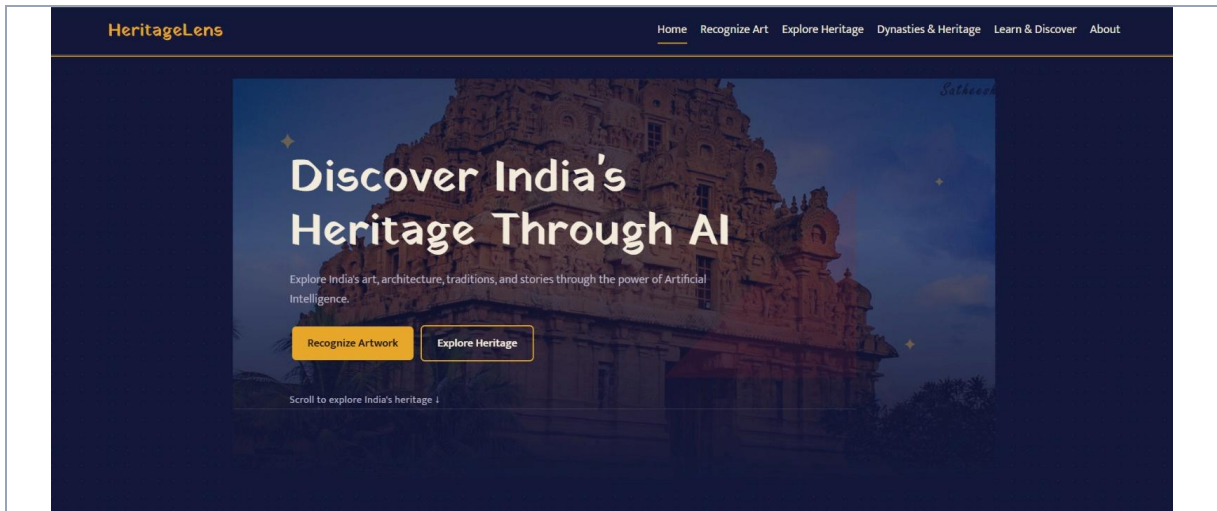


Figure 1: Home page of HeritageLens

10.2 Recognize Art Page

The Recognize Art page contains the upload area, the Analyze image button, and the progress stages. When an image is selected, a preview is displayed inside the upload area.

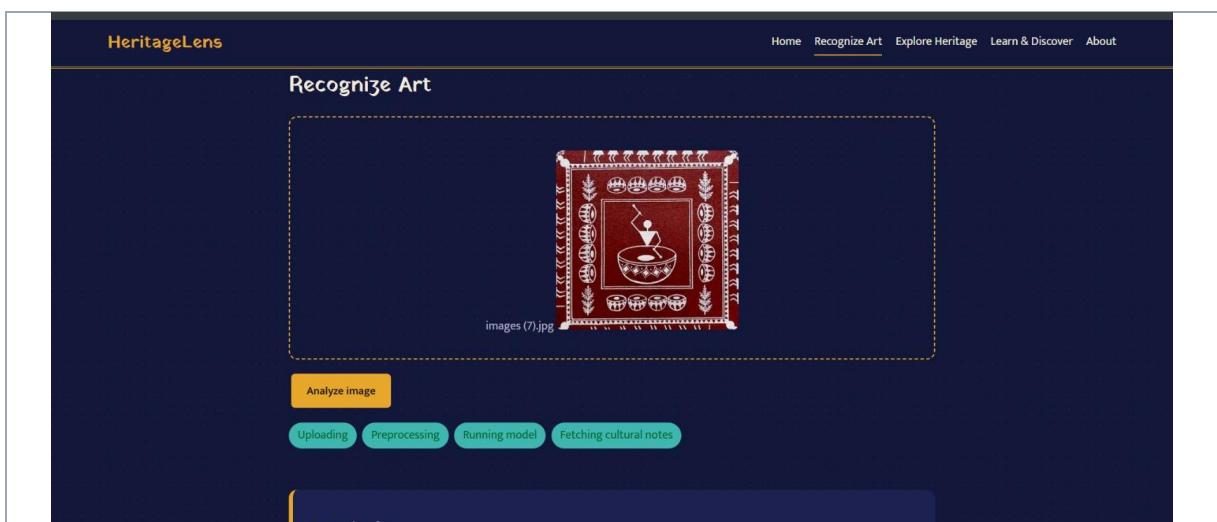


Figure 2: Recognize Art page with an uploaded Warli artwork

10.3 Recognition Result

After analysis, the result panel shows the detected art form (Warli), the confidence (61.1%), and the other candidates (Mandana Art 13.1% and Kalamkari 11.9%). Below this, the cultural profile is organized into Origin, History, Cultural significance, Characteristics, Materials, Traditional techniques, and Interesting facts.

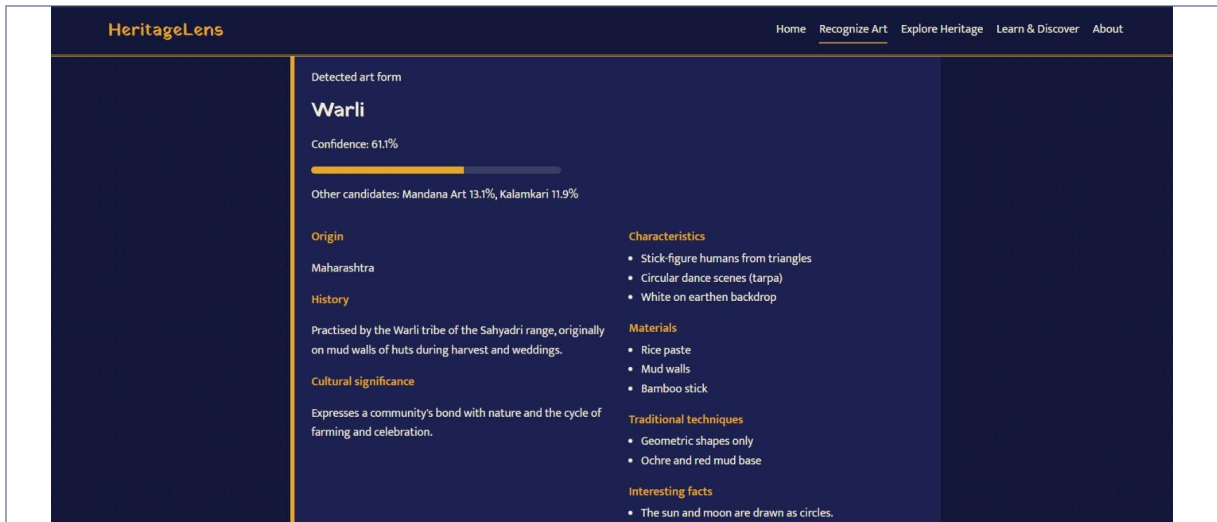


Figure 3: Recognition result with confidence and cultural information

10.4 Recent Recognitions

Below the upload area, recent recognitions are shown as cards containing the thumbnail, the detected art form, and the confidence. In the example, Tanjore Painting was recognized with 96.9% and 97.7% confidence and Warli with 61.1%.

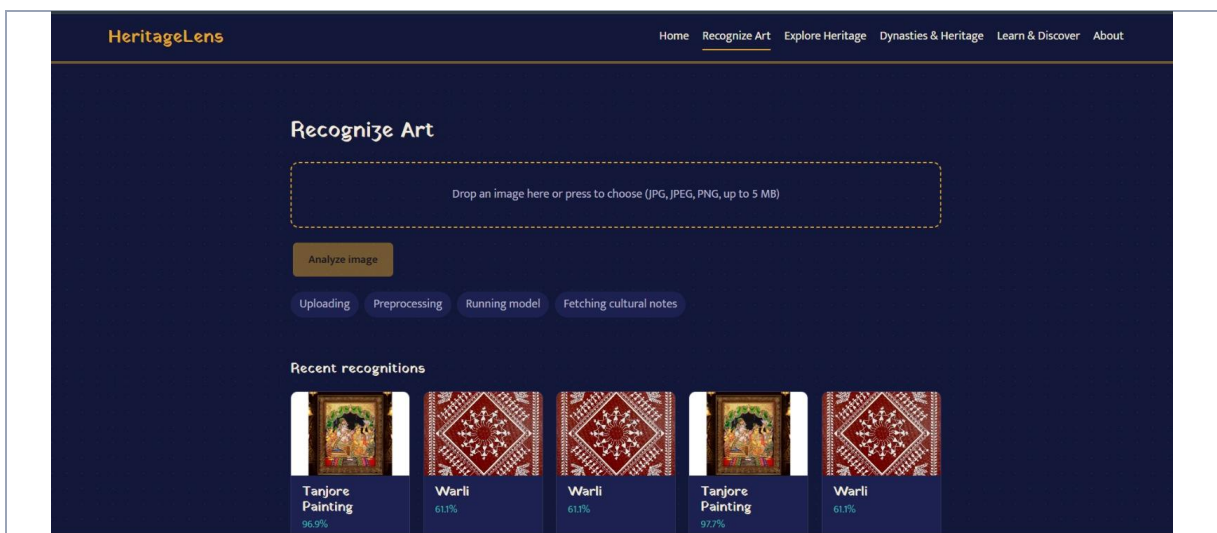


Figure 4: Recognize Art page showing recent recognitions

10.5 Explore Heritage

The Explore Heritage page displays art forms as cards with an image, name, region, and short description. Users can search by name and filter by region and technique.

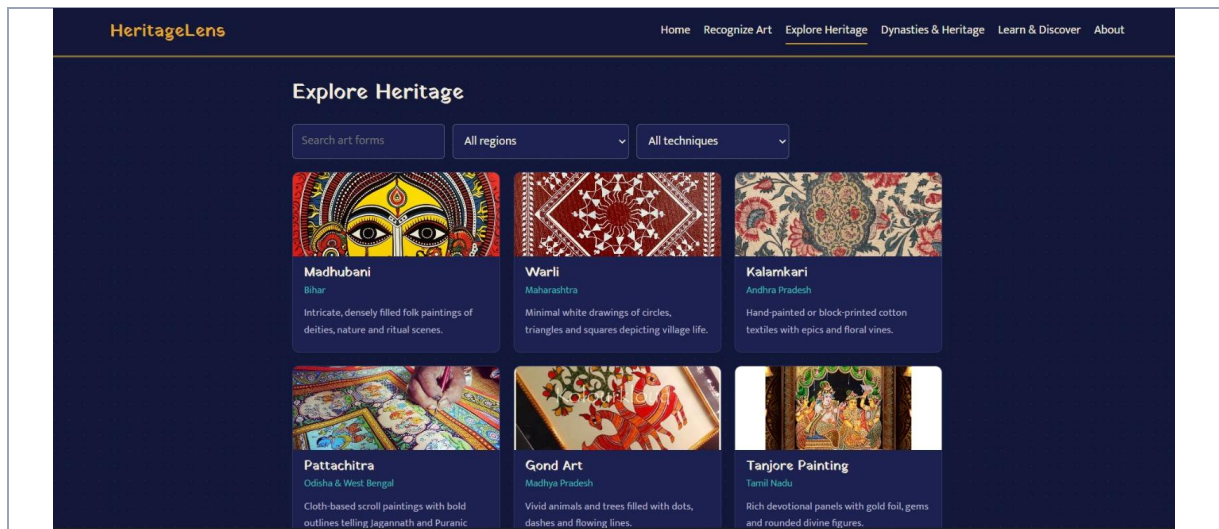


Figure 5: Explore Heritage gallery with search and filters

10.6 Dynasties and Heritage

The Dynasties and Heritage page shows cards for the Maurya, Gupta, Pallava, Chalukya, Rashtrakuta, and Chola dynasties. Each card lists the period and the region, and users can search or filter by era.

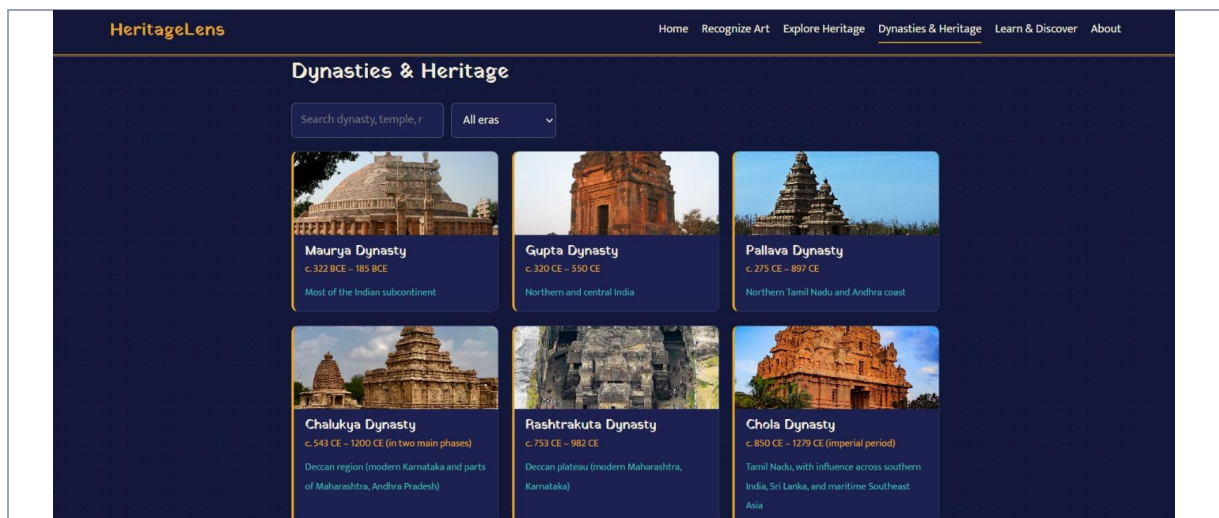


Figure 6: Dynasties and Heritage page

11. Testing and Validation

Testing verifies that each feature behaves as specified. The table below lists the functional test cases together with their expected results.

Table 11: Functional test cases

No.	Test case	Expected result
1	Upload a valid JPG image	Prediction, confidence, candidates, and cultural information are displayed
2	Upload a valid PNG image	Same as above
3	Upload an unsupported file (e.g., PDF)	An error message is shown; no prediction is made
4	Upload an image larger than 5 MB	An error message about the size limit is shown
5	Click Analyze image without selecting a file	The user is asked to choose an image first
6	Recognize several images	Entries appear in Recent recognitions with correct labels
7	Search for "Warli" in Explore Heritage	Only the matching card is displayed
8	Filter Explore Heritage by region or technique	Only art forms that match the filter are displayed
9	Search or filter by era in Dynasties and Heritage	Only matching dynasties are displayed
10	Open the application on a phone or tablet	The layout adapts and remains readable
11	Start the server without a model file	The application runs in demo mode with a clear label
12	Start the server with a trained model	<code>/api/status</code> reports real mode and predictions come from the model

Model quality is validated separately from these functional tests. It should be measured on a held-out test set using the evaluation measures described in Section 8.7, and the results should be reported alongside this document.

12. Security, Ethics, and Limitations

12.1 Security and Error Handling

- Only JPG, JPEG, and PNG files up to 5 MB are accepted.
- Uploaded files are opened with Pillow; corrupt or invalid images produce a handled error message rather than a crash.
- No API keys or sensitive information are stored in the source code.
- The recognition history is stored only in the user's own browser.

12.2 Ethical and Cultural Considerations

Cultural content must be presented with care and accuracy. The knowledge base should be reviewed against reliable sources and, where possible, by cultural experts and practising artists. The application makes clear that predictions are educational guidance and should not be treated as authoritative authentication. Demo predictions are always labelled so that users are never misled about the source of a result.

12.3 Limitations

- The accuracy of the model depends on the quality, size, and variety of the training dataset.
- Visually similar styles can be confused, which is why the confidence score and alternative candidates are shown.
- The model can only choose from the ten trained classes, so an image of another art form will still receive the closest match.
- The recognition history is lost if the user clears browser data.
- The cultural information is summarized and cannot capture the full depth of each tradition.

13. Future Scope

1. **More art forms:** extend the model and knowledge base to additional regional traditions such as Kalighat, Tikuli, Miniature, and Sanjhi styles.
2. **Unknown-image handling:** add a confidence threshold so that images outside the supported classes are reported as not recognized.
3. **Visual explanations:** use techniques such as Grad-CAM to highlight the image regions that influenced a prediction.
4. **Expanded dynasties:** add further dynasties and individual monuments, with maps and timelines.
5. **Multilingual support:** provide the interface and cultural content in Indian languages.
6. **User accounts and cloud history:** store history in a database such as SQLite or PostgreSQL to sync across devices.
7. **Mobile and camera support:** deliver a mobile application with live camera recognition, for use at museums and heritage sites.
8. **Expert verification and collaboration:** work with cultural organizations and artists to review, enrich, and validate the content.

14. Conclusion

HeritageLens demonstrates how Artificial Intelligence and Computer Vision can serve cultural education and heritage preservation. By combining a MobileNetV2-based CNN with a Flask REST API, a JSON knowledge base, and a responsive heritage-themed interface, the system recognizes traditional Indian art forms and presents their history, significance, materials, and techniques. The Dynasties and Heritage section connects this artistic knowledge with the historical rulers and monuments that shaped India's cultural landscape.

The modular design allows new art forms, dynasties, and improved models to be added over time, making the project a scalable foundation for further work. With a well-trained model and expert-reviewed content, HeritageLens can become a valuable resource for students, tourists, researchers, and cultural organizations.

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